

Smart City Complaint Analysis

1. Introduction

Smart cities receive thousands of citizen complaints every day regarding public services such as garbage collection, water supply, road maintenance, street light failures, drainage issues, traffic congestion, and public safety. Managing and resolving these complaints efficiently is essential for improving urban living standards.

This project aims to analyze smart city complaint data using data analytics techniques and develop dashboards that help city authorities identify complaint patterns, monitor service performance, and improve citizen satisfaction.

2. Problem Statement

City authorities are experiencing an increasing number of citizen complaints. Monitoring complaint trends manually is difficult and may result in delayed resolutions and reduced citizen satisfaction.

The objective of this project is to identify complaint patterns, analyze service performance, detect complaint hotspot zones, and provide recommendations that improve city operations.

3. Objectives

- To analyze complaint patterns across different city zones.

- To identify the most frequent complaint categories.
- To evaluate complaint resolution performance.
- To investigate the relationship between complaint priority and resolution delay.
- To identify complaint hotspot areas.
- To assess department efficiency and citizen satisfaction.
- To provide recommendations for improving city services.

4. Dataset Description

The dataset contains information related to citizen complaints registered across different city zones. **Main Attributes**

- Complaint_ID
- Complaint_Date
- Resolved_Date
- Category
- Department
- Zone
- Priority
- Status
- Citizen_Satisfaction

The dataset was analyzed using Python, Pandas, NumPy, Matplotlib, Seaborn, and Power BI.

Phase 1: Data Understanding & Cleaning

The Smart City Complaint dataset was loaded and examined to understand its structure and quality. Missing values were handled, duplicate records were removed, and date columns were converted into proper datetime format. Complaint categories and text fields were cleaned for consistency. Additionally, derived variables such as Complaint Severity Score and Resolution Delay were created to support further analysis.

Phase 2: KPI Design

In this phase, key performance indicators (KPIs) are defined to measure business performance and success. KPIs are selected based on project objectives and data availability. They help track important metrics such as efficiency, performance, trends, and outcomes. Proper KPI design ensures that the analysis focuses on meaningful insights and supports better decision-making.

Phase 3: Analytics Investigation

Relationships between complaint characteristics and operational performance were analyzed.

Zone vs Complaint Count

Complaint volumes were compared across different city zones to identify recurring problem areas.

Category vs Resolution Time

Complaint categories were analyzed against average resolution delay to determine which issues take longer to resolve.

Department vs Performance

Departments were evaluated based on complaint volume and resolution efficiency.

Complaint Trends Over Time

Monthly complaint trends were analyzed to identify seasonal patterns and complaint spikes.

Priority vs Resolution Delay

The impact of complaint priority on resolution performance was examined.

Visualizations

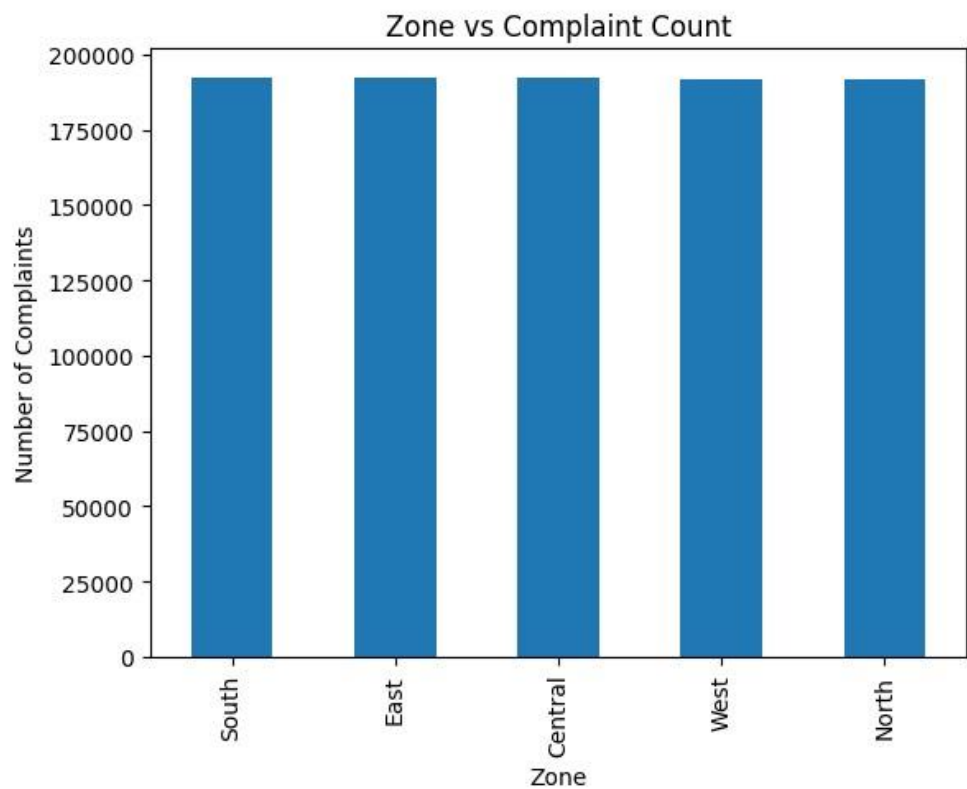
- KPI Cards
- Line Charts
- Clustered Bar Charts
- Stacked Bar Charts

- Heatmaps
- Geographic Maps

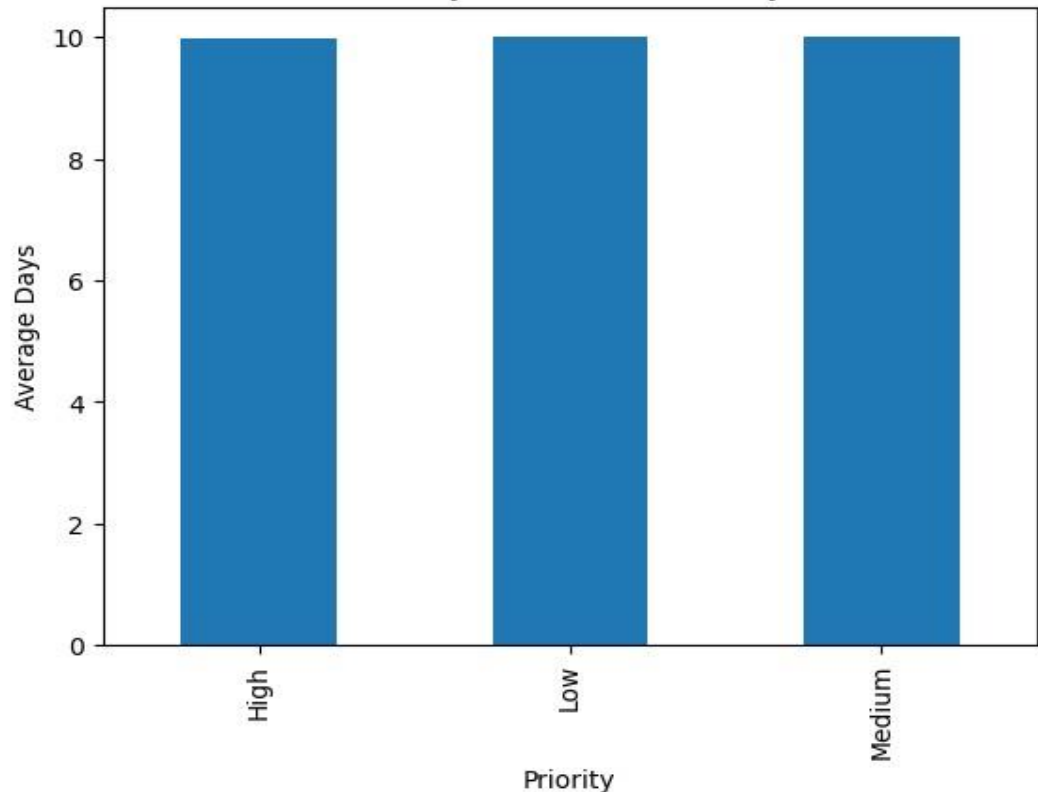
These visualizations helped identify patterns and operational bottlenecks.

Visualizations:

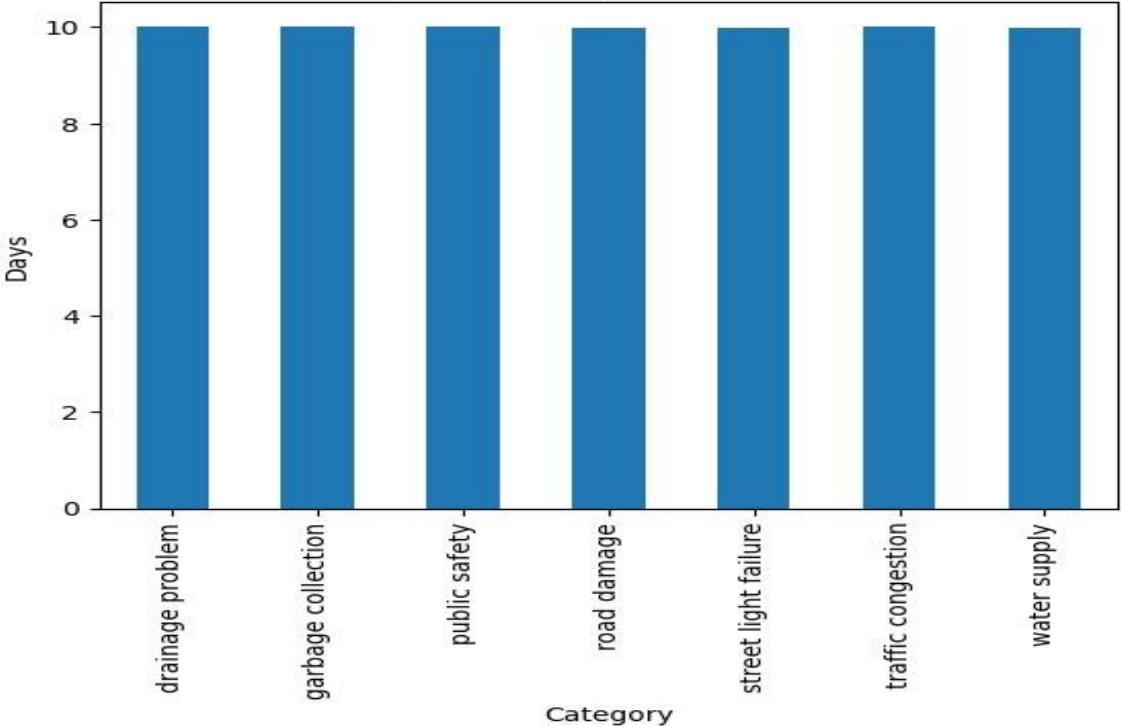
1)Bar chart



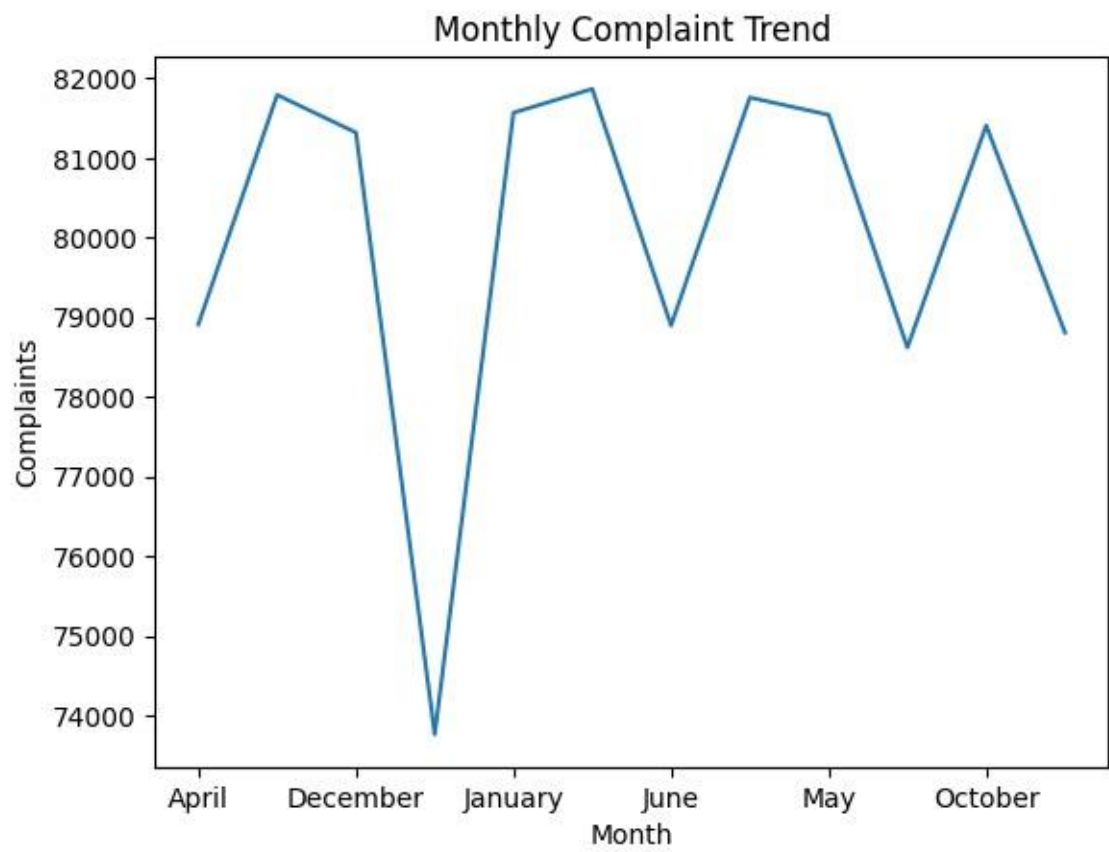
Priority vs Resolution Delay



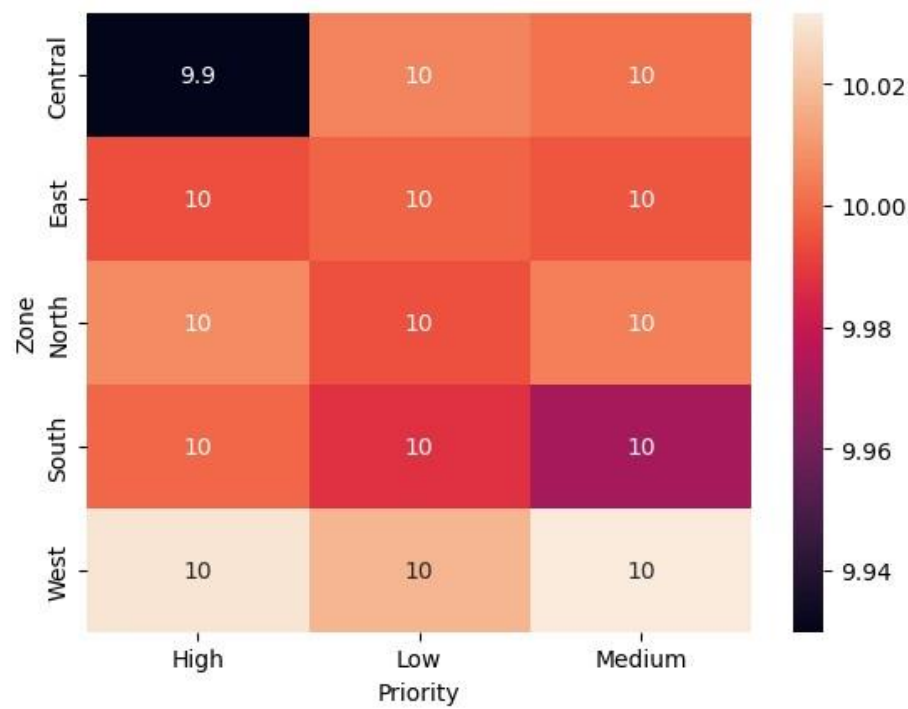
Category vs Average Resolution Time



2) Linechart



3) Heatmap



Phase 4: Dashboard Development

Executive Overview Dashboard

The Executive Dashboard provides a high-level summary of city operations. Included Metrics:

- Total Complaints
- Resolution Rate
- Citizen Satisfaction Score
- Complaint Trends

The dashboard enables leadership teams to monitor city performance quickly.

Complaint Analysis Dashboard

The Complaint Dashboard focuses on complaint behaviour and category analysis. Analysis Performed:

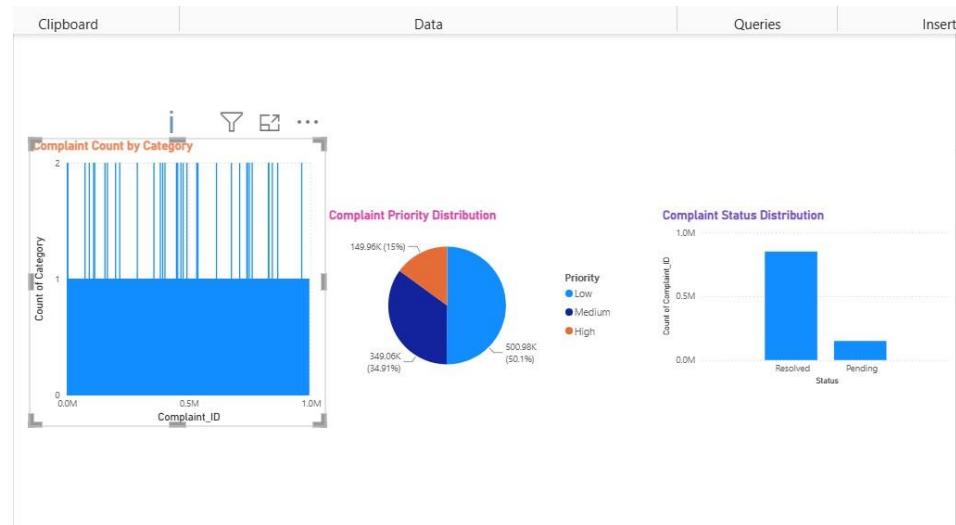
- Complaint Categories
- Resolution Patterns
- Complaint Growth Trends Insights

Generated:

- Most frequent complaint categories
- Complaint status distribution

- Category-wise complaint trends

Visualization:



Geographic Dashboard

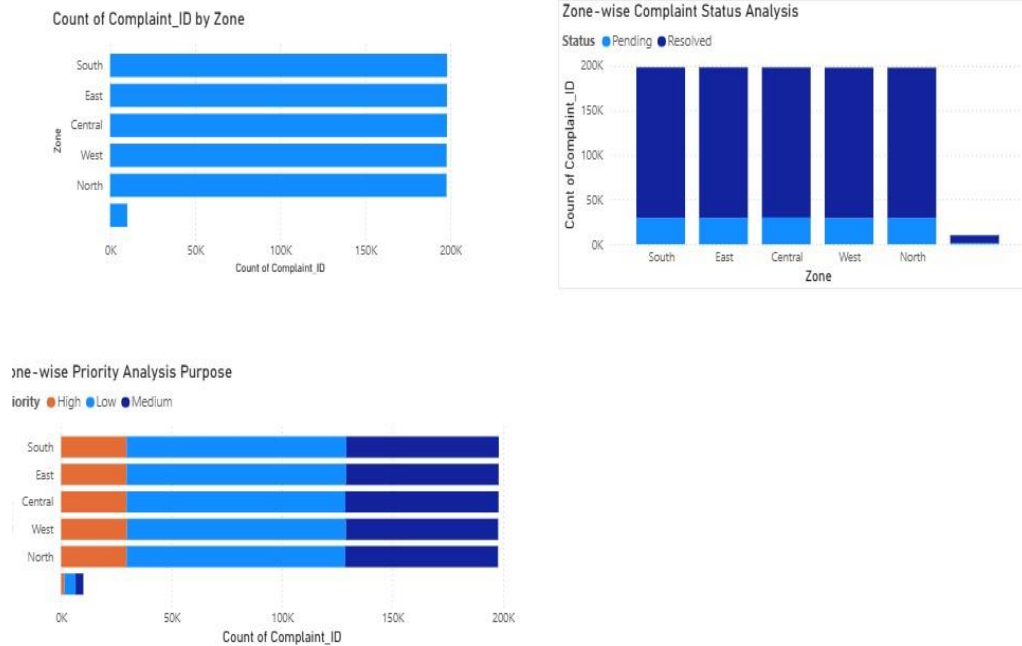
The Geographic Dashboard visualizes complaint distribution across city zones. Analysis Performed:

- Zone-wise Issues
- Complaint Hotspots
- Geographic Distribution Insights

Generated:

- High complaint zones
- Recurring problem areas
- Regional service issues

Visualization:



Service Performance Dashboard

The Service Performance Dashboard evaluates operational efficiency.

Analysis Performed:

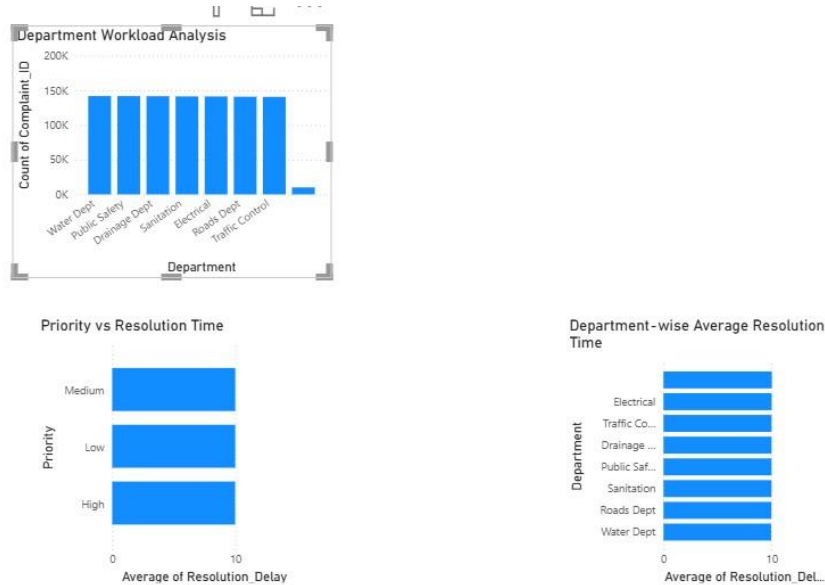
- Department Performance
- Resolution Efficiency
- Priority Complaint Handling Insights

Generated:

- Best-performing departments
- Departments with long resolution times

- Department workload comparison

Visualization:



Business Insights

1. Which complaint categories occur most frequently?

- Traffic Congestion generated the highest number of complaints.
- Street Light Failures and Drainage Problems were also frequently reported.
- Water Supply and Public Safety complaints contributed significantly to the total complaint volume.
- These categories require continuous monitoring and faster response mechanisms.

2. Which areas experience recurring civic problems?

- The South Zone(1,98,143)recorded the highest complaint volume, indicating recurring issues related to civic services and infrastructure.
- This zone should be prioritized for maintenance activities, resource allocation, and service improvements.
- Although the difference between zones is relatively small, South remains the primary complaint hotspot in the city.

3. Which departments take the longest to resolve complaints?

- The Electrical Department, Traffic Control Department, and Public Safety Department take the longest time to resolve complaints, with an average resolution time of approximately 10 days.
- The longer resolution periods may be due to the complexity of infrastructure-related issues and higher complaint volumes.

4. What factors reduce citizen satisfaction?

- Long complaint resolution times significantly reduce citizen satisfaction.
- Unresolved and pending complaints negatively affect public trust.
- Repeated complaints from the same zone indicate ineffective problem resolution.
- Poor service quality and delayed responses lead to lower satisfaction ratings.

5. What actions should city authorities prioritize?

- Reduce complaint resolution delays through better resource allocation.
- Focus on high-priority complaints for faster service delivery.
- Improve infrastructure in complaint hotspot zones.
- Strengthen department accountability and performance monitoring.
- Implement real-time complaint tracking and citizen feedback systems.

Recommendations Operational Improvements

- Reduce complaint resolution time.
- Improve resource allocation in hotspot zones.
- Strengthen complaint monitoring systems.
- Implement escalation mechanisms for high-priority complaints.

Service Management

- Increase preventive maintenance activities.
- Improve traffic management systems.
- Upgrade drainage and street lighting infrastructure.

Citizen Engagement

- Improve communication with citizens.
- Collect regular citizen feedback.
- Introduce real-time complaint tracking systems.

Conclusion

This project transformed citizen complaint data into meaningful insights using data analytics and interactive dashboards. The analysis highlighted key problem areas, service gaps, and resolution patterns across the city. The developed dashboards can support faster decision-making, improve public service management, and contribute to a more efficient and responsive smart city system.